2nd semester problem sets

June 8, 2026

1 Problem set 1

Q1. Determine which of the following are subrings of $\mathbb{Q}$

a. rational numbers with odd denominator

b. rational numbers with even denominator

c. nonnegative rationals

d. squares of rationals

e. rationals with odd numerator

f. rationals with even numerator

Soln: Let's check option (a). Let $\mathbb{Q}' = \left\{ \frac{a}{b} \mid a \in \mathbb{Z}, b = 2k+1, \text{for some integer } k \right\}$ Suppose $\frac{a}{b}$ and $\frac{c}{d}$ are two elements of the set $\mathbb{Q}'$. Then we have that $b = 2k+1$ and $d = 2m+1$ for some integers k, m. Now,

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Since an odd integer times an odd integer is always odd, we get that $\frac{a}{b} - \frac{c}{d} \in \mathbb{Q}'$. Also, clearly, $1 \in \mathbb{Q}'$and $0 \in \mathbb{Q}'$. So, $\mathbb{Q}'$ is a subring of $\mathbb{Q}$ dgjldikjfgldsjg;oekgsoj **Q2**